

Name: _____

Period: _____

Polynomials IV — Multiplicity, Symmetry, End Behavior, Sketching & Binomial

“Even multiplicity *boing!* (bounce). Odd multiplicity *whoosh!* (cross).”

Quick Reference

Multiplicity: If $(x - r)^m$ divides $p(x)$, then r is a root of multiplicity m .

Behavior at a root: m odd $\Rightarrow$ crosses the x -axis; m even $\Rightarrow$ bounces.

Even/odd functions: f even if $f(-x) = f(x)$ (y-axis symmetry). f odd if $f(-x) = -f(x)$ (origin symmetry).

End behavior: Leading term ax^n :

n even: both ends same ($a > 0$ up, $a < 0$ down). n odd: ends opposite ($a > 0$ left down/right up; $a < 0$ left up/right down).

Max extrema: $\leq n - 1$. **Max inflection points:** $\leq n - 2$.

Binomial theorem: $(a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$.

1. (6 points) **Roots & multiplicities.**

For $p(x) = (x - 4)^2(x + 1)^3(x - 2)$:

- (a) List all real roots with their multiplicities.

- (b) State whether the graph crosses or bounces at each root.

2. (6 points) **Construct from multiplicities.**

Build a *monic* polynomial with real roots -2 (double), 0 (single), and 3 (triple). Write it factored.

3. (6 points) **Minimum degree from behavior.**

A polynomial has x -intercepts at $x = -1$ (bounce), $x = 2$ (cross), and no other real intercepts. What is the *smallest* possible degree? Explain briefly.

4. (6 points) **Even/odd/neither.**

Classify each as even, odd, or neither. Justify with $f(-x)$.

(a) $f(x) = x^5 - x^3$

(b) $g(x) = x^4 - 2x^2 + 1$

(c) $h(x) = x^3 + x^2$

5. (6 points) **Design an even polynomial.**

Find a degree-4 *even* polynomial with real roots ± 3 and y -intercept 18. (Hint: even $\Rightarrow$ only even powers.)

6. (6 points) **End behavior from leading term.**

Describe the end behavior (as $x \rightarrow \pm\infty$) for each:

(a) $p(x) = -5x^7 + \dots$

(b) $q(x) = 2x^6 + \dots$

(c) $r(x) = -x^4 + \dots$

7. (6 points) **Infer leading sign/degree.**

A polynomial goes *down* on the far left and *up* on the far right. What can you conclude about the parity of the degree and the sign of the leading coefficient?

8. (6 points) **Interpreting multiplicity locally.**

At $x = 1$, a graph touches the x -axis and turns upward; near $x = 1$ the function looks like $(x - 1)^m$ for some m . Is m even or odd? Explain.

9. (6 points) **Intercept check.**

For $f(x) = (x - 2)^2(x + 4)$, find all x -intercepts with bounce/cross labels, and the y -intercept.

10. (6 points) **Max counts from degree.**

If $\deg P = 7$, what are the maximum possible numbers of relative extrema and inflection points?

11. (8 points) **Rough sketch from data (no graphing tech).**

Given $P(x) = (x + 2)^2(x - 1)^3$:

- (a) Degree and leading term (up to constant).

- (b) End behavior (left/right).

- (c) x -intercepts with multiplicities and bounce/cross.

- (d) y -intercept.

12. (8 points) **Rough sketch 2.**

For $Q(x) = -2(x - 4)^2(x + 1)$, answer the same four items as in Q11.

13. (6 points) **Derivative hint about repeated roots (concept).**

Suppose $x = 2$ is a root of multiplicity 3 of $p(x)$. Which of $p(2), p'(2), p''(2)$ must be 0? (No formal proof required.)

14. (6 points) **Pick the multiplicity.**

At $x = -3$ the graph *crosses* the axis but flattens (touches with a little “S” shape). Which multiplicity is most consistent: 1, 2, or 3? Explain briefly.

15. (6 points) **Leading behavior from factor form.**

For $R(x) = (x - 5)^4(2x + 1)$, determine the degree, sign of leading coefficient, and end behavior.

16. (6 points) **Count real roots from a sketch brief.**

A degree-6 polynomial has x -intercepts at -4 (double), -1 (single), and 3 (double), plus one more real root. What is the multiplicity of the remaining root?

17. (6 points) **Binomial basics (examples).**
Use the binomial theorem to expand $(x + 2)^3$.

18. (6 points) **A binomial coefficient hunt.**
Find the coefficient of x^4 in $(2x - 3)^6$.

19. (6 points) **Middle term idea.**
What is the middle term in the expansion of $(x + 1)^8$? (Write the full term.)

20. (10 points) **Full feature summary.**
Let $S(x) = -x(x + 1)^2(x - 3)^2$. List degree, leading term sign, end behavior, all intercepts with multiplicities and cross/bounce, and the y -intercept.

